2022-2023 SPRING SEMESTER FLUID MECHANICS FINAL EXAM, Date:05/06/2023

Name: **Surname:** **Student ID:**_ _ _ _ _ _ _ _ _
b a

QUESTION 2 (25p): The water taken from the reservoir is discharged to the atmosphere by the pipe system shown in the figure. If incompressible and ideal fluid are assumed:

a) What is the flow rate of the system?

b) What are the velocities in each pipe?

c) What is the smallest possible diameter of the pipe (2) without vaporization for this calculated flow rate of the system?

The water temperature is 20 C°, and the atmospheric pressure at sea level is 101.325 kPa.

$D_1=D_3=0.4a$ meter, $D_2=0.2b$ meter



